

An illustration of a classroom scene. In the foreground and middle ground, several students of diverse backgrounds are seated at their desks, facing towards the right. They are wearing casual clothing like hoodies and sweaters. In the background, a large network diagram is projected onto the wall. The diagram consists of several circular nodes, some containing icons of people, connected by a web of lines. A clock is also visible on the wall to the left of the network diagram. The overall lighting is soft and the colors are muted, creating a calm academic atmosphere.

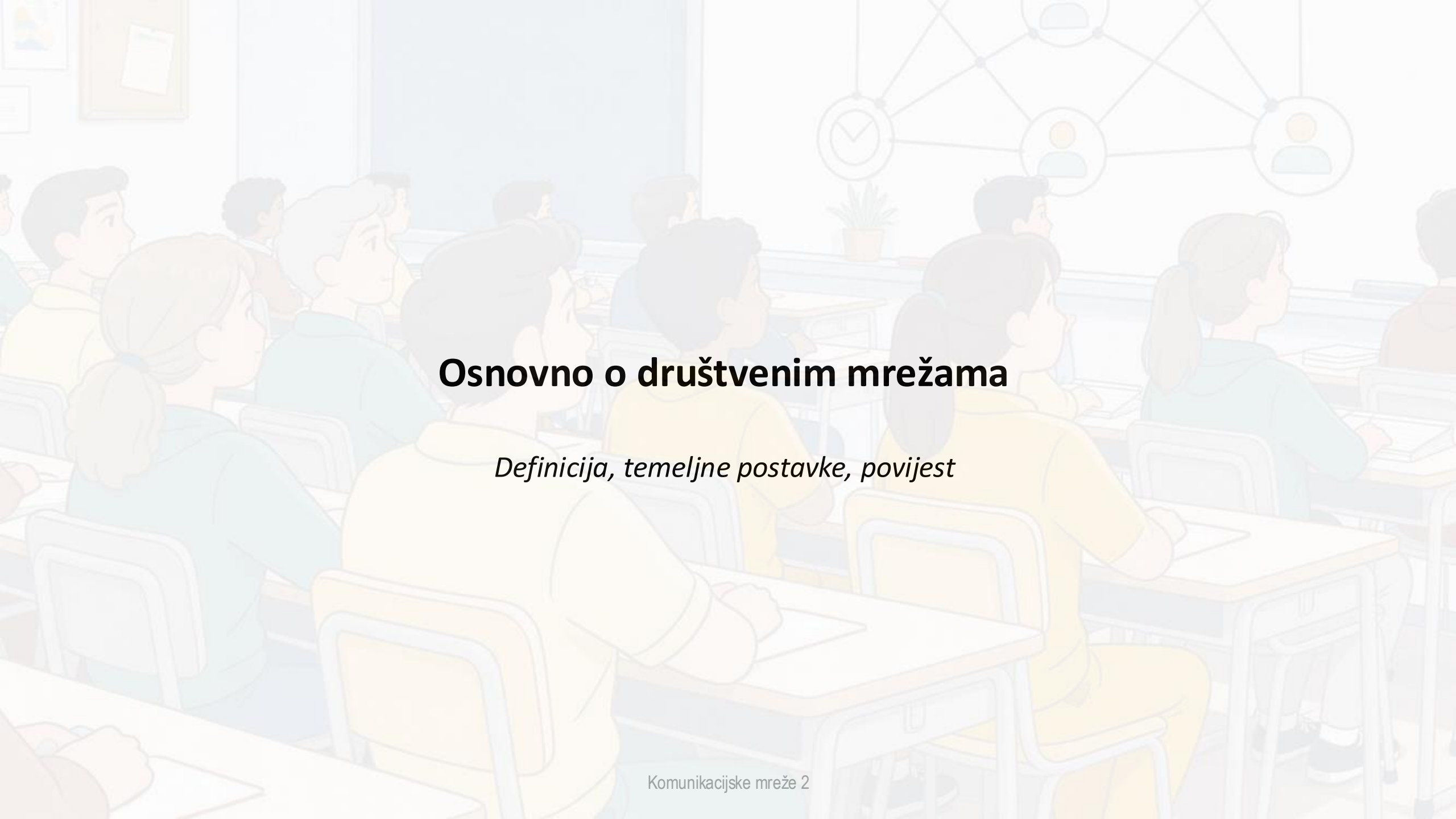
Komunikacijske mreže 2

14. predavanje

Studijski slučaj: Društvene mreže

Sadržaj

- Osnovno o društvenim mrežama
- (Rana) evolucija društvenih mreža zasnovanih na internetskoj infrastrukturi
- Društvene mreže danas: trendovi i statistika
- Primjeri ICT-usluga zasnovanih na društvenim mrežama



Osnovno o društvenim mrežama

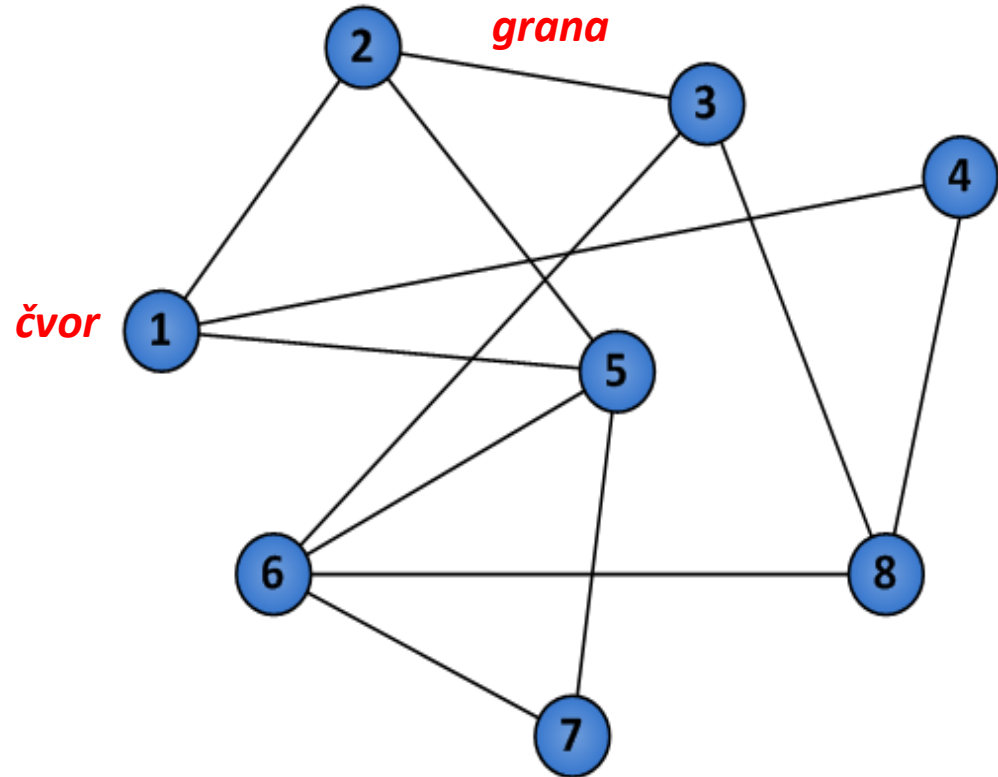
Definicija, temeljne postavke, povijest

Što je to mreža?



Što je to mreža?

- Skup **čvorova** međusobno povezanih **granama**

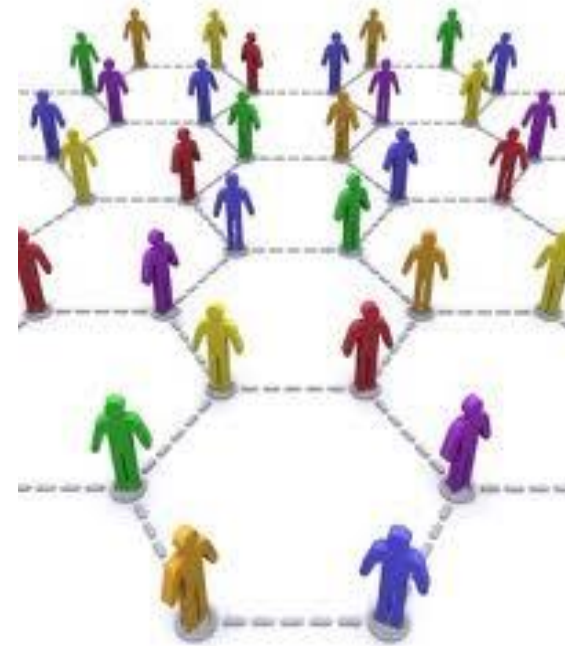


Što je to društvena mreža?



Što je to društvena mreža?

- Skup **entiteta** međusobno povezanih **odnosima**
 - Entiteti: *ljudi, grupe ljude, organizacije, brandovi*
 - Odnosi: *poznanstvo, obiteljska povezanost, privlačnost, ...*

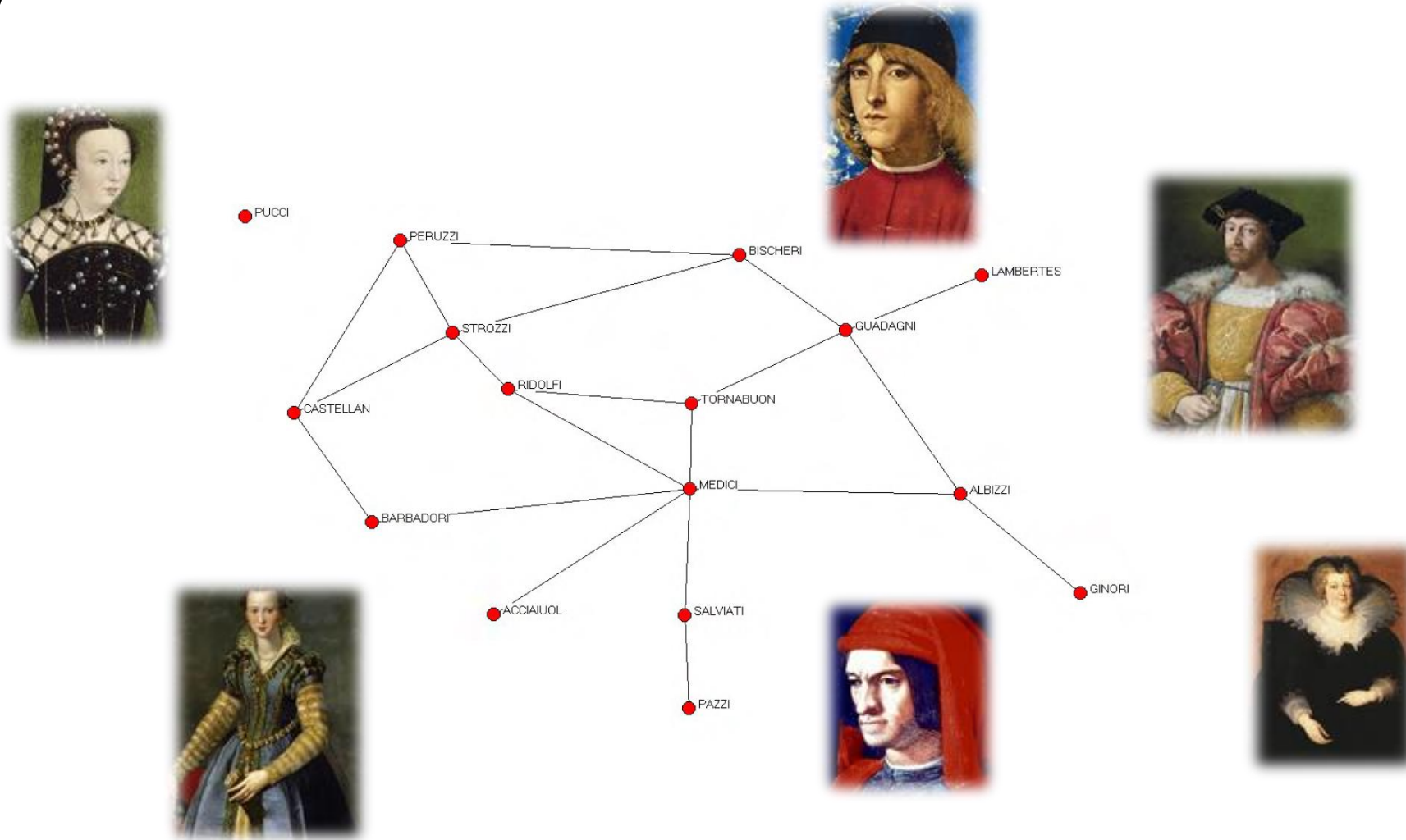


The hidden influence of social networks

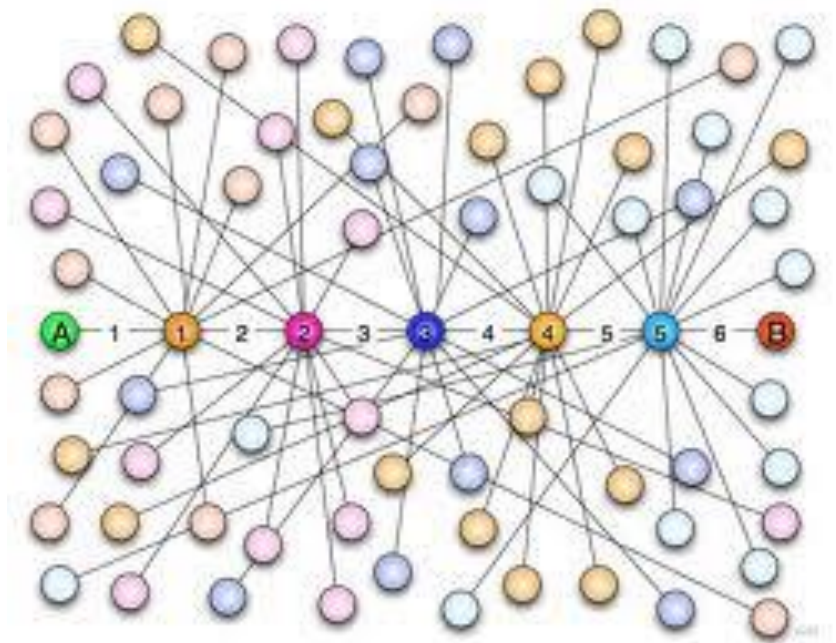
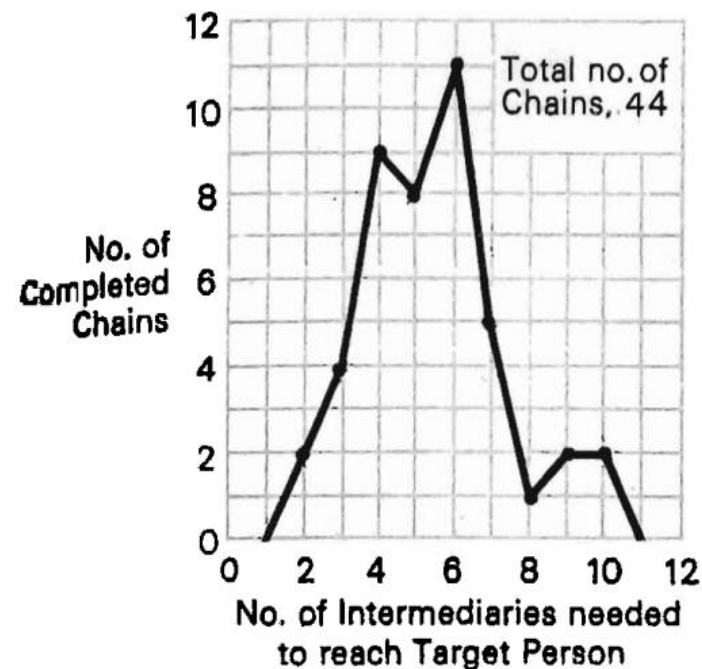
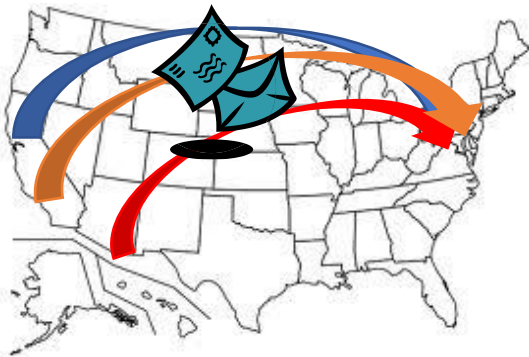


[The hidden influence of social networks](#)

Povijest društvenih mreža (1): *Utjecaj obitelji Medici* (1430)

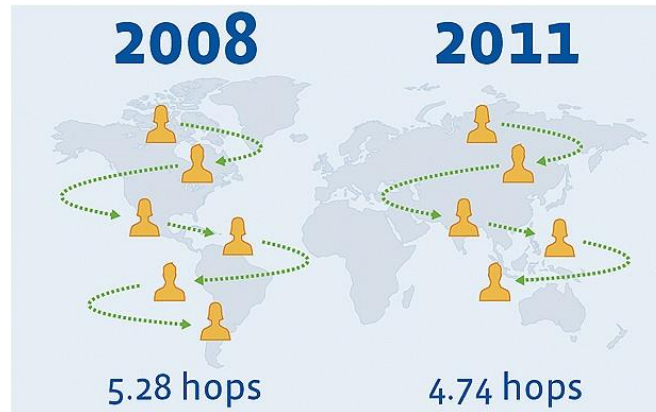


Povijest društvenih mreža (2): *Milgramov „small world” eksperiment: „six degrees of separation”*



Six degrees of separation u Facebook eri

facebook



2016
3,57

Tipovi društvenih mreža

- Društvene mreže se mogu različito klasificirati
- **Eksplicitne društvene mreže**
 - **Poznati odnosi** među aktorima
 - Facebook, Twitter, LinkedIn i Instagram su dobri primjeri eksplicitnih društvenih mreža



- **Implicitne društvene mreže**
 - **Poveznice među aktorima** mogu biti utvrđene samo **dubljom analizom** dostupnih podataka
 - Amazon ili Netflix
 - Analizom logova mogu se zaključiti zajednički interesi poput političkih naklonosti ili hobija



**(Rana) evolucija društvenih mreža zasnovanih na
internetskoj infrastrukturi**

1985. – 2008.

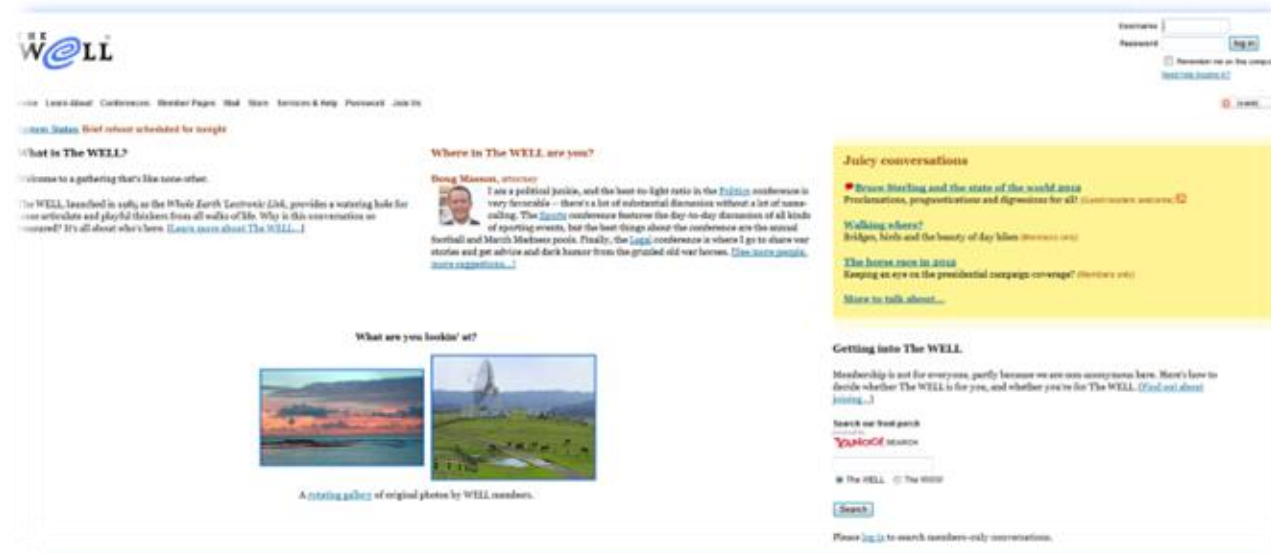
Koja je najstarija društvena mreža i kada je osnovana?



Evolucija društvenih mreža zasnovanih na Internetu (1)

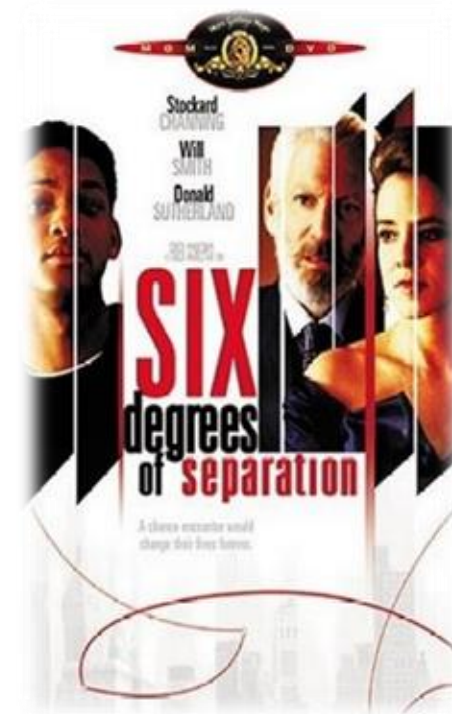


Evolucija društvenih mreža zasnovanih na Internetu (2): Društvene mreže prije WWW



Evolucija društvenih mreža zasnovanih na Internetu (3): Prva društvena mreža u razdoblju Web-a 1.0

- SixDegrees.com (1997 - 2000)
- Teorija „six degrees of separation”
- Profili, prijatelji, liste prijatelja



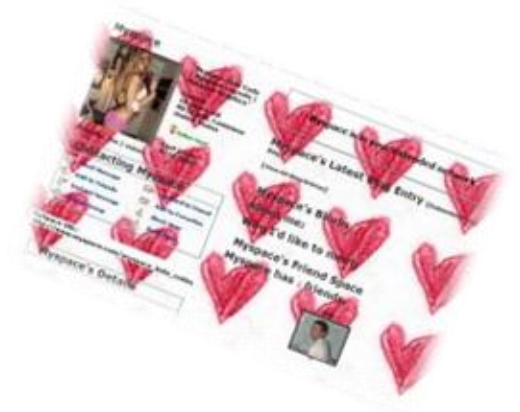
Evolucija društvenih mreža zasnovanih na Internetu (4): Društvene mreže u razdoblju Web-a 2.0

- 4 kategorije:
 - **Zabava**
 - Spajanje s ljudima radi druženja i zabave
 - SixDegrees.com, Friendster
 - **Posao**
 - Širenje poslovne mreže
 - LinkedIn, Plaxo, VisiblePath, Xing
 - **Razmjena sadržaja**
 - Djeljenje Web sadržaja sa ostalim korisnicima
 - Flickr, Last.FM, Youtube
 - **Mreže specifične domene**
 - Spajanje korisnika sličnih interesa
 - Dogster, Catster, Care2, MyChurch



Evolucija društvenih mreža zasnovanih na Internetu (6): MySpace (2003.)

- „Efekt mreže” osam mjeseci nakon lansiranja
- Ime „MySpace” proizlazi od opcije korisničkog uređivanja HTML koda
- Više od 80 milijuna korisnika u 2006.
- Veliki sigurnosni problemi



Evolucija društvenih mreža zasnovanih na Internetu (7): Facebook (2004.)

- 2004. Mark Zuckerberg skupa sa tri kolege lansira web stranicu Thefacebook.com
- Registracija zahtjeva harvard.edu e-mail domenu, ostali korisnici mogli su sudjelovati tek u kasnijim fazama
- U rujnu 2005. Thefacebook mijenja ime u Facebook



Evolucija društvenih mreža zasnovanih na Internetu (8): Facebook vs MySpace (>100M korisnika)

		
Verifikacija korisnika	Ima verifikaciju	Nema verifikaciju
Baza korisnika	Vrhunski studenti u SAD-u	SAD
Dostupnost	Ekskluzivna mreža („urban elite“)	Dostupno svima
Reklame	Nema reklama	Puno reklama
Dizajn	Moderno	Kaotično

Evolucija društvenih mreža zasnovanih na Internetu (9)



- Facebook vs. MySpace (otvorena vs. zatvorena platforma)
 - MySpace nema otvoren API
- Facebook ima otvoren API
 - Nezavisne kompanije razvijaju Facebook aplikacije
 - Slide, RockYou, zynga

slide®



zynga



Facebook Launches Facebook Platform; They are the Anti-MySpace

MICHAEL ARRINGTON

Thursday, May 24th, 2007

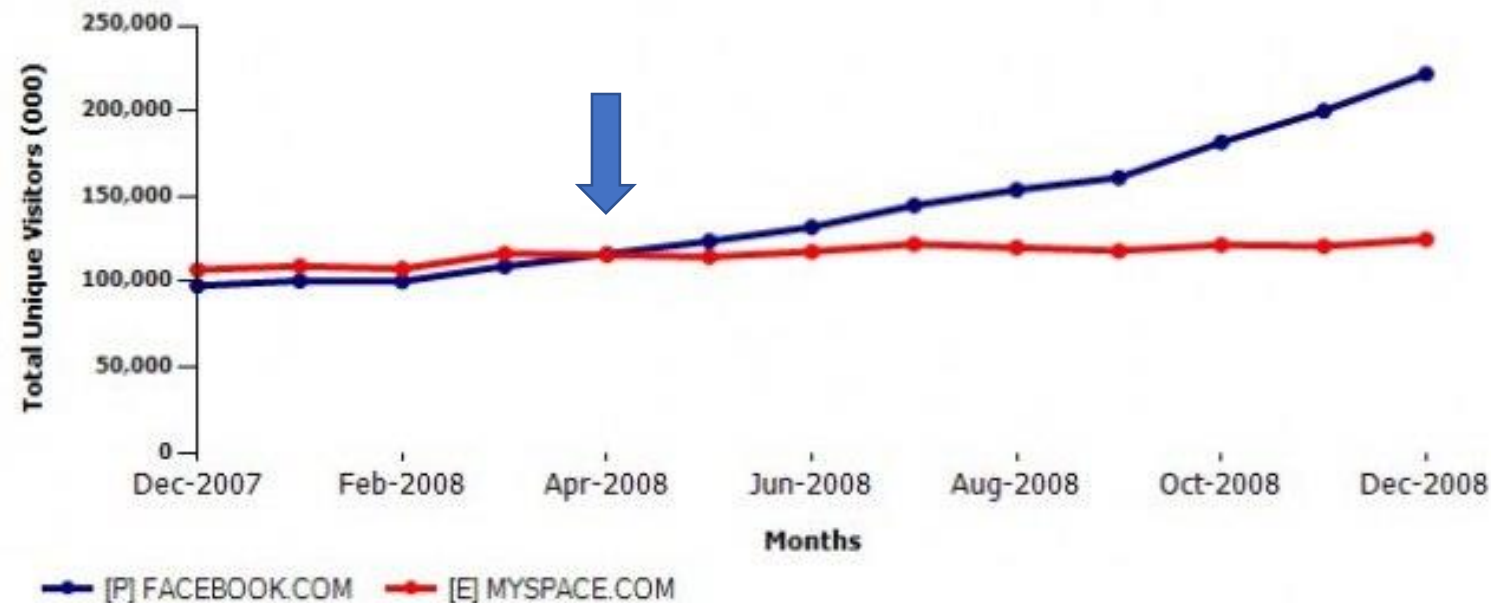
1 Comments

Facebook is holding a massive press/developer event today in San Francisco to officially launch Facebook Platform. 750 or so people are here.

A number of third party applications will also be announced, including Microsoft, Amazon, Slide, RockYou, Box.net, Red Bull, Washington Post, Project Anane, Prosnor, Snarvine, ilike, PicksPal, Digg, Plum and others. Seventy

facebook

Evolucija društvenih mreža zasnovanih na Internetu (10)



The background of the slide is a light-colored illustration of a classroom. Students are seated at desks, facing forward. On the wall, there is a clock and a network diagram consisting of several circular nodes connected by lines. Some nodes contain icons of people. The overall style is clean and modern.

Društvene mreže danas: trendovi i statistika

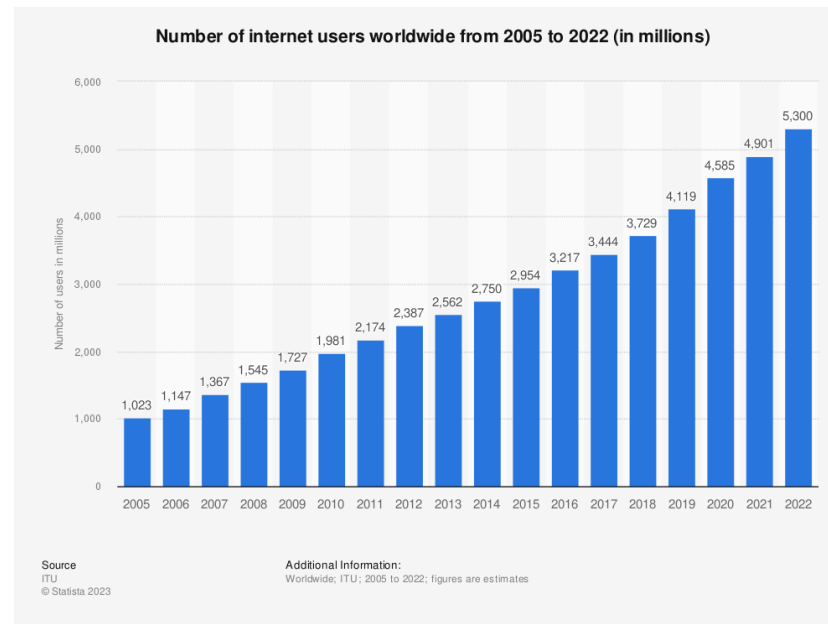
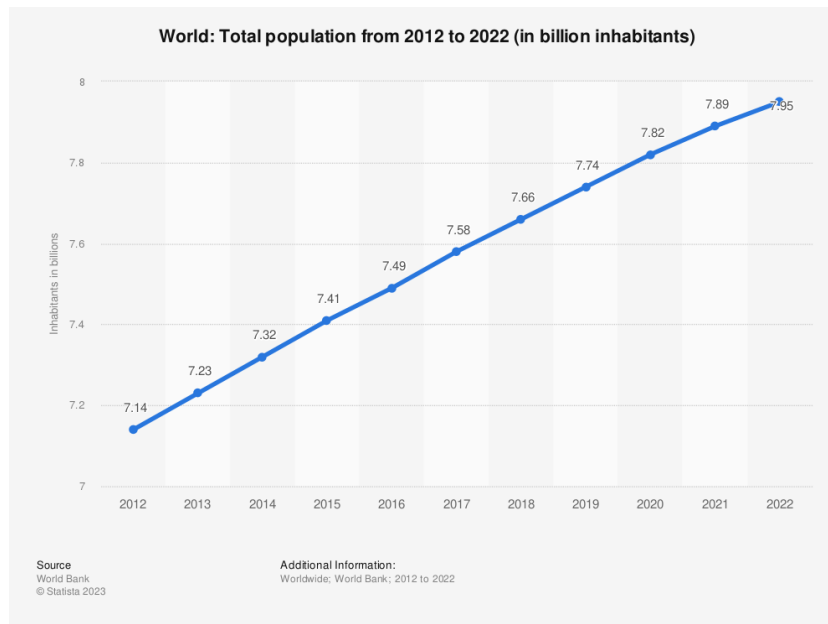
Ljudi vs korisnici Interneta



Ljudi vs korisnici Interneta



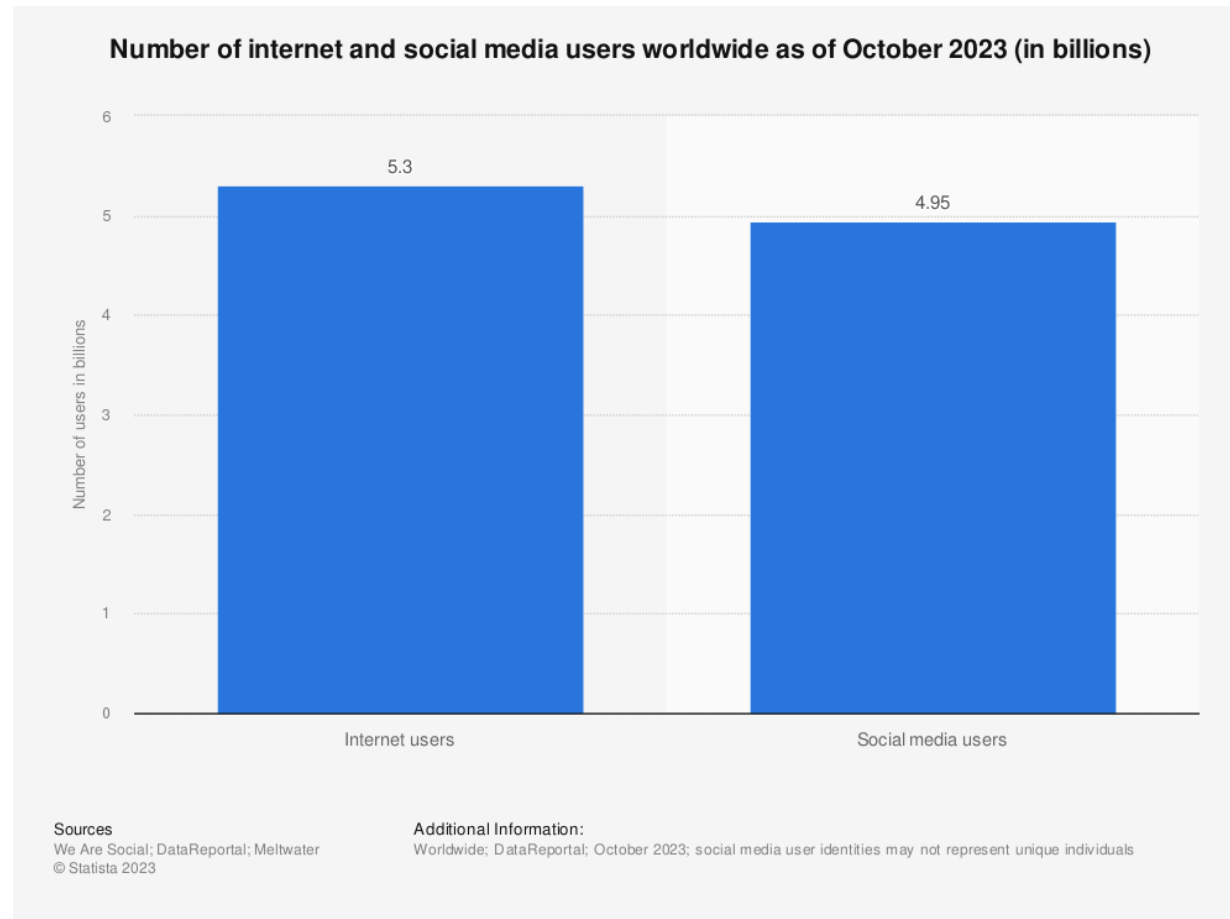
67%



Korisnici Interneta vs korisnici društvenih mreža

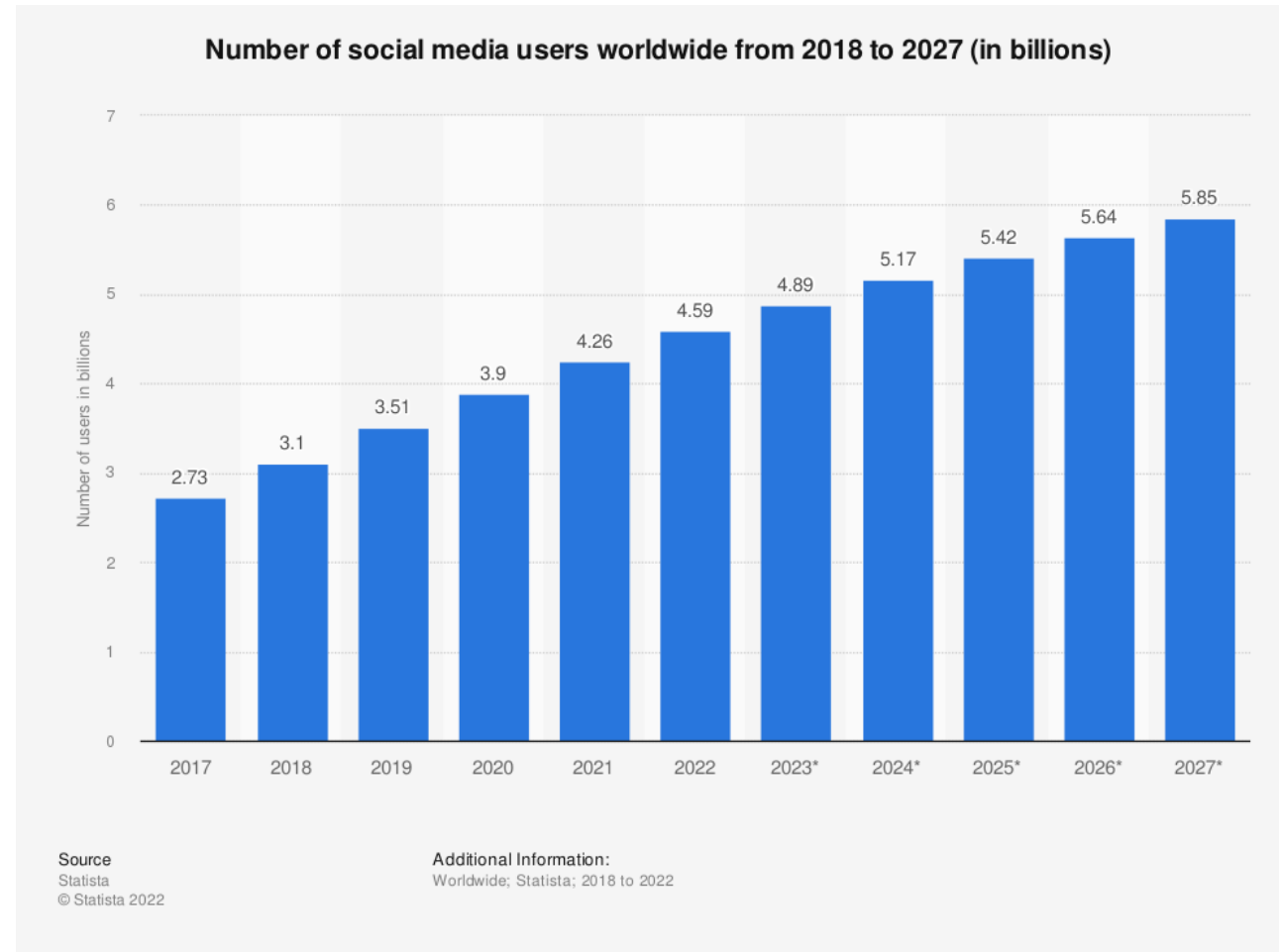


Korisnici Interneta vs korisnici društvenih mreža



93%

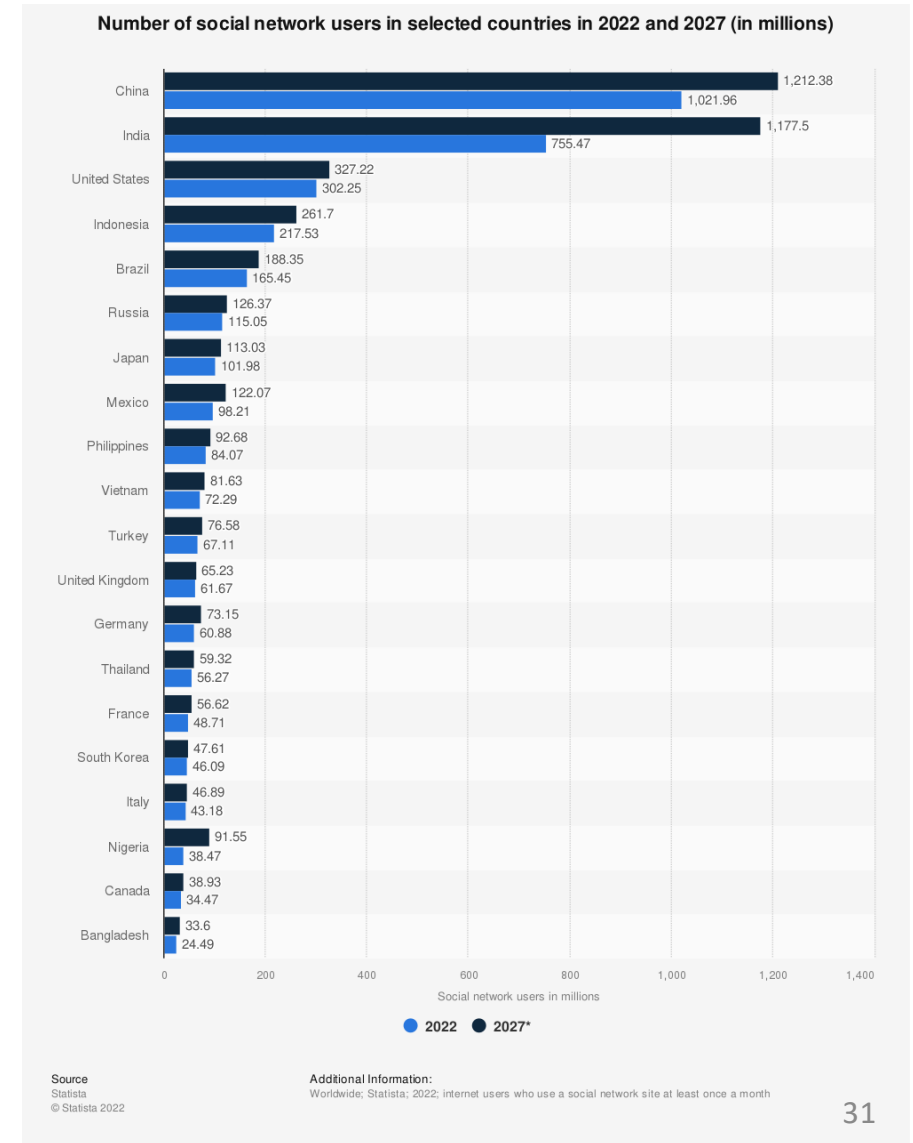
Projekcija rasta broja korisnika društvenih mreža



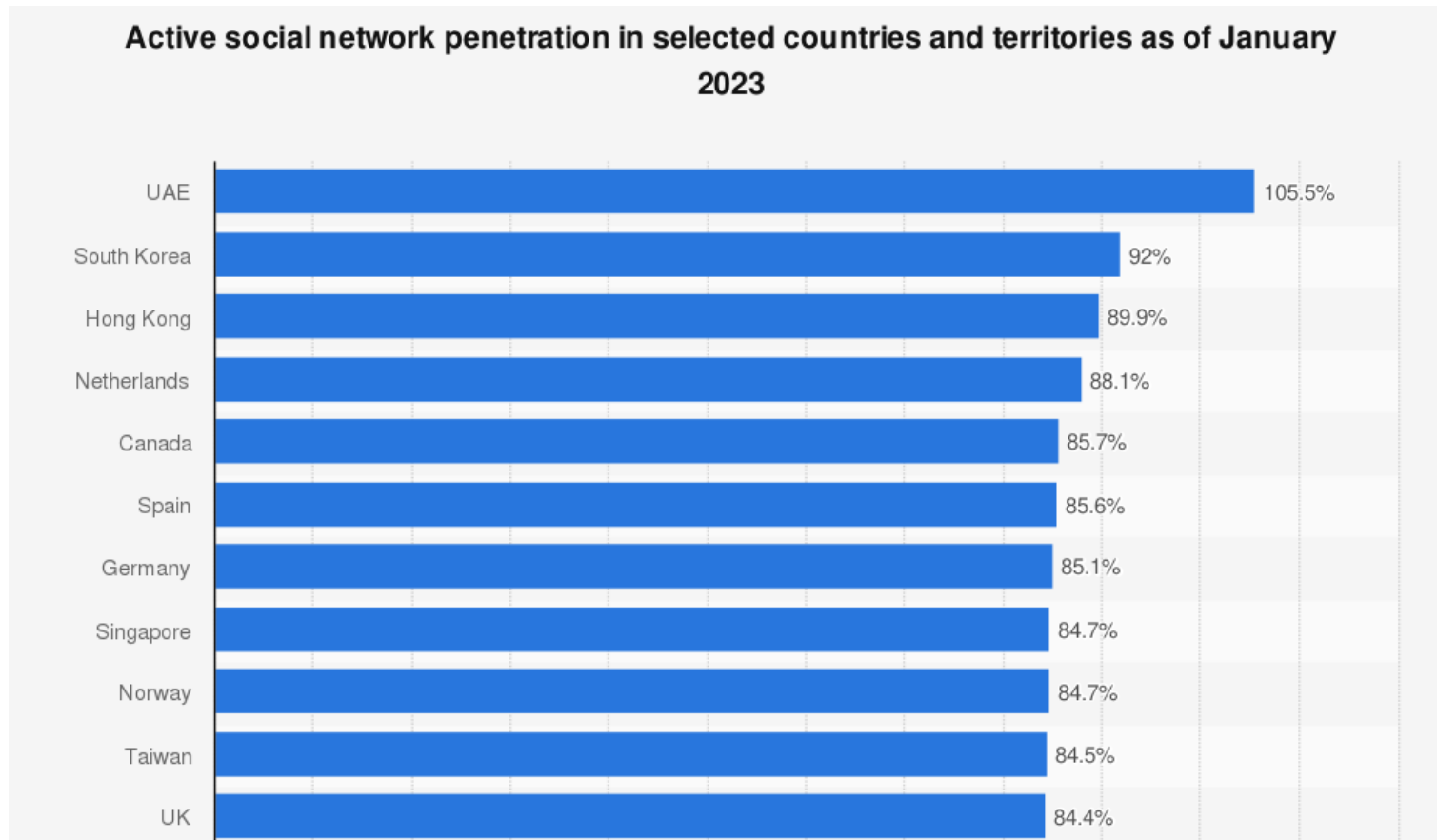
Distribucija korisnika društvenih mreža po državama (apsolutni broj) – top 3 zemlje



Distribucija korisnika društvenih mreža po državama (apsolutni broj)

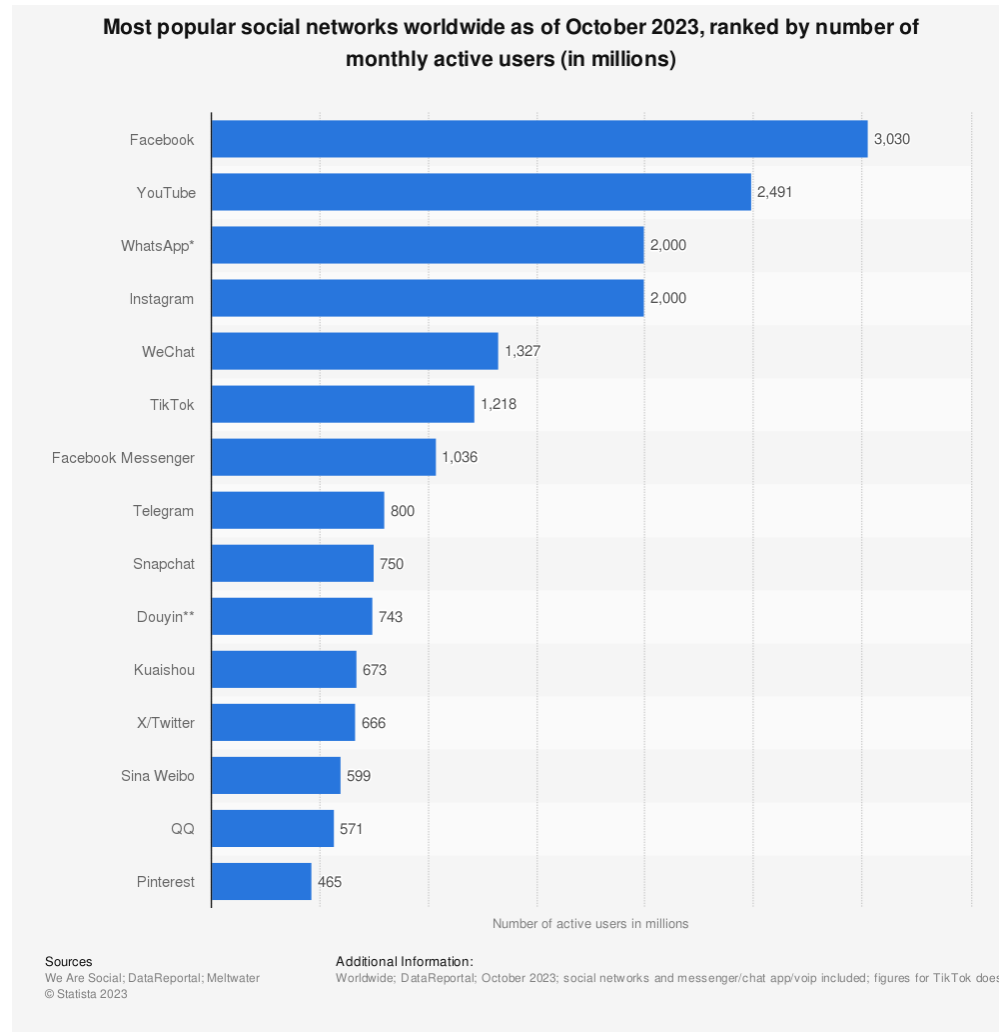


Distribucija korisnika društvenih mreža po državama (relativno vs populacija)



WW=59%

Najpopularnije društvene mreže



LinkedIn
310M

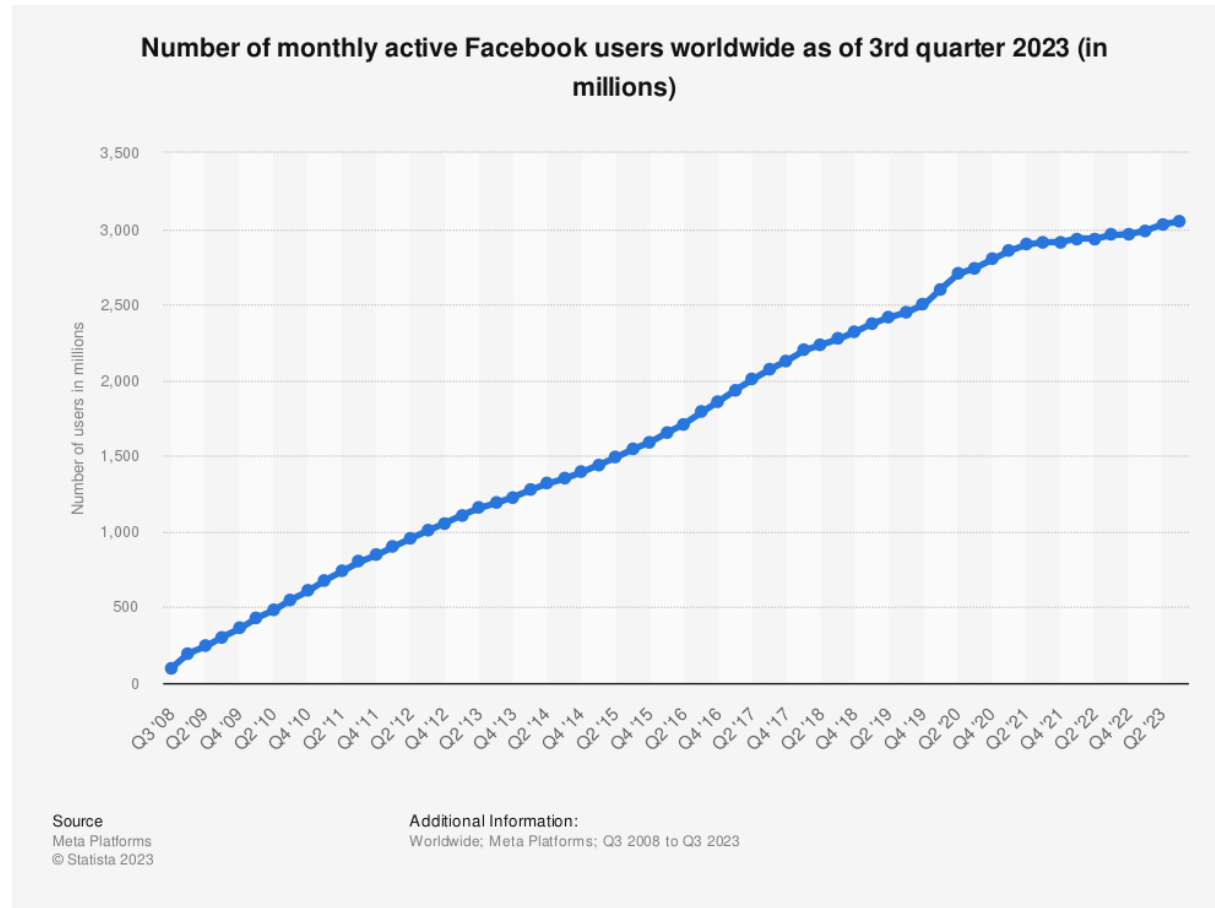


Facebook – broj korisnika WW & Hrvatska





Facebook rast broja korisnika (mjesečno aktivni)



CRO
1,9M

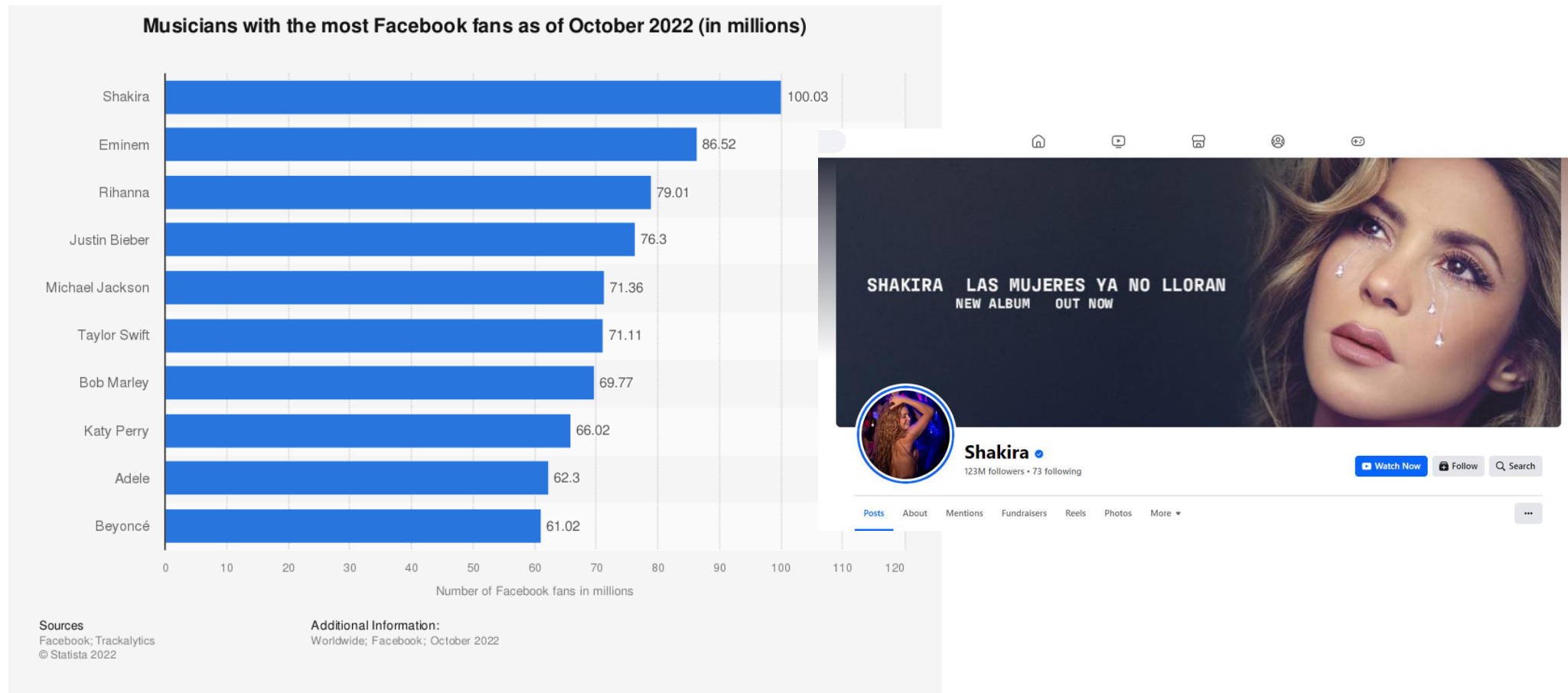


Facebook – najpopularnija umjetnica

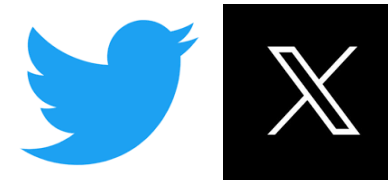




Facebook – najpopularnija umjetnica

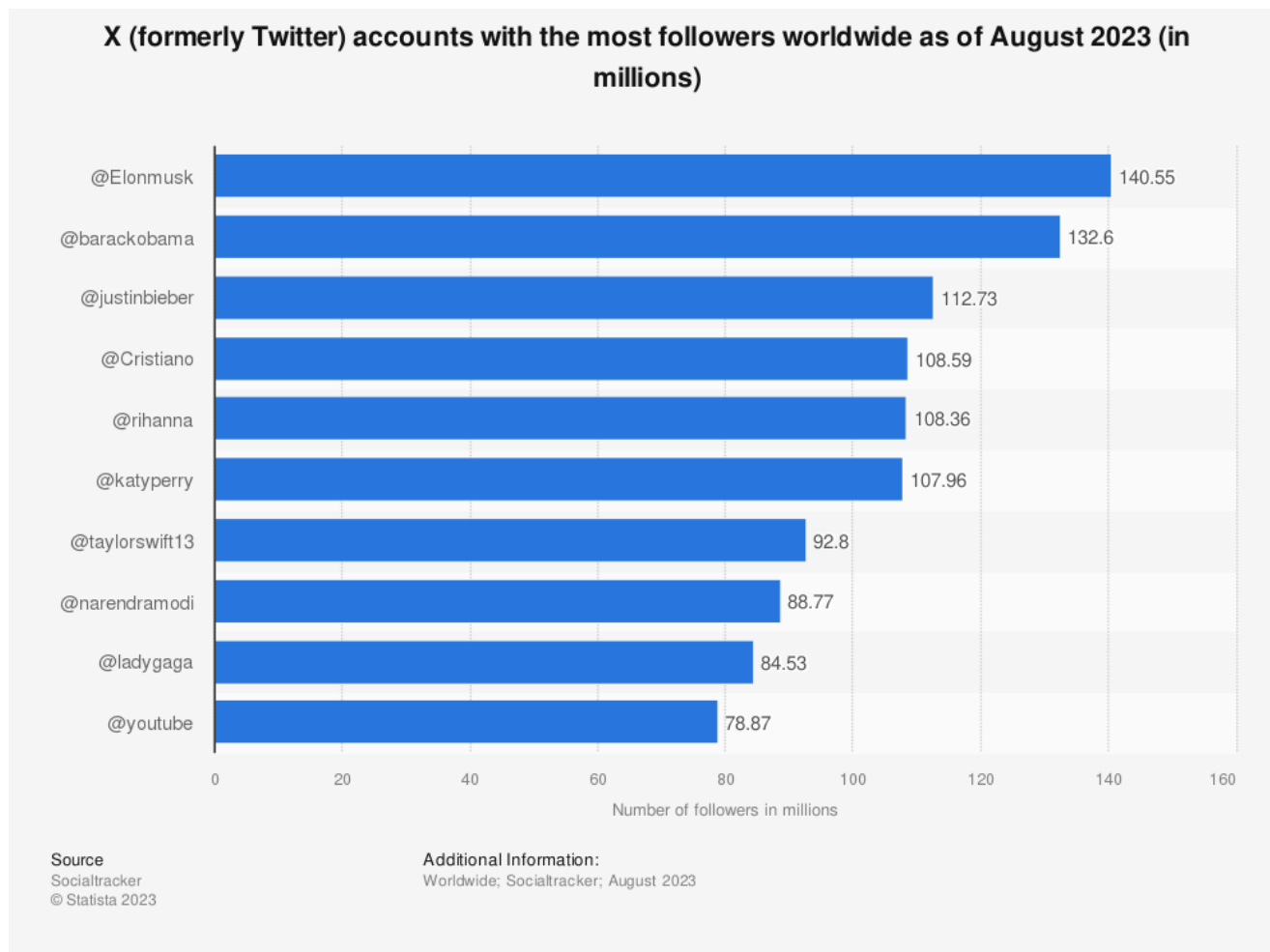


Twitter / X – najpopularniji profil





Twitter / X – najpopularniji profili





Twitter / X & Elon Musk (love & hate)

2022.



2023.



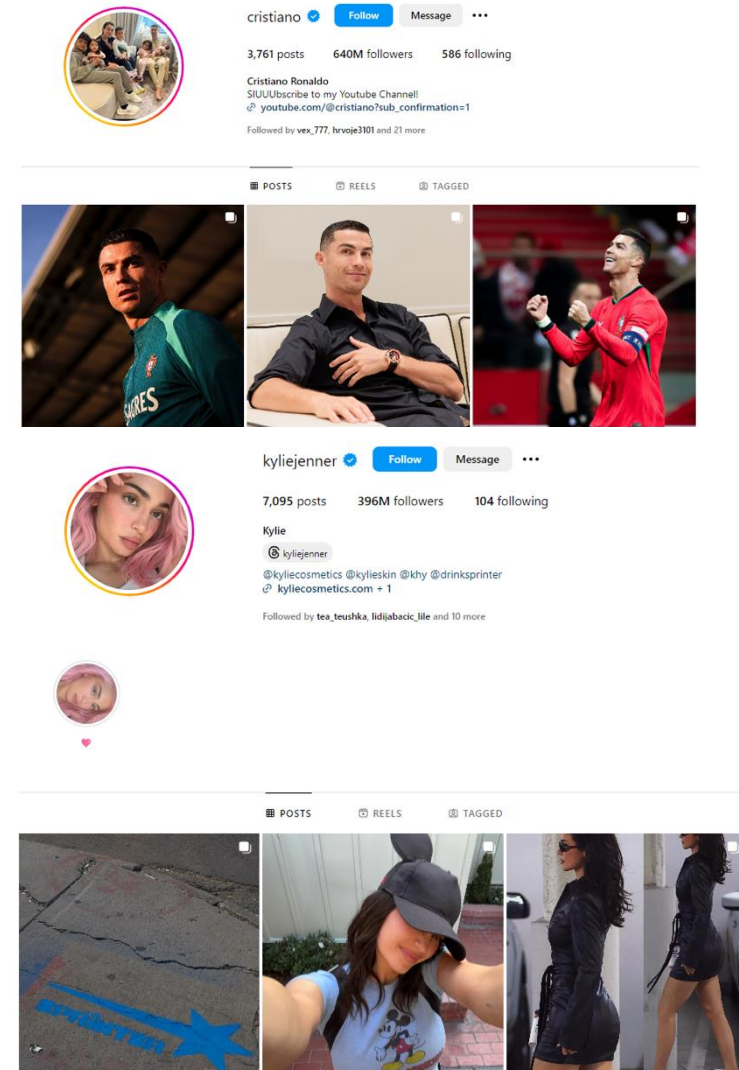
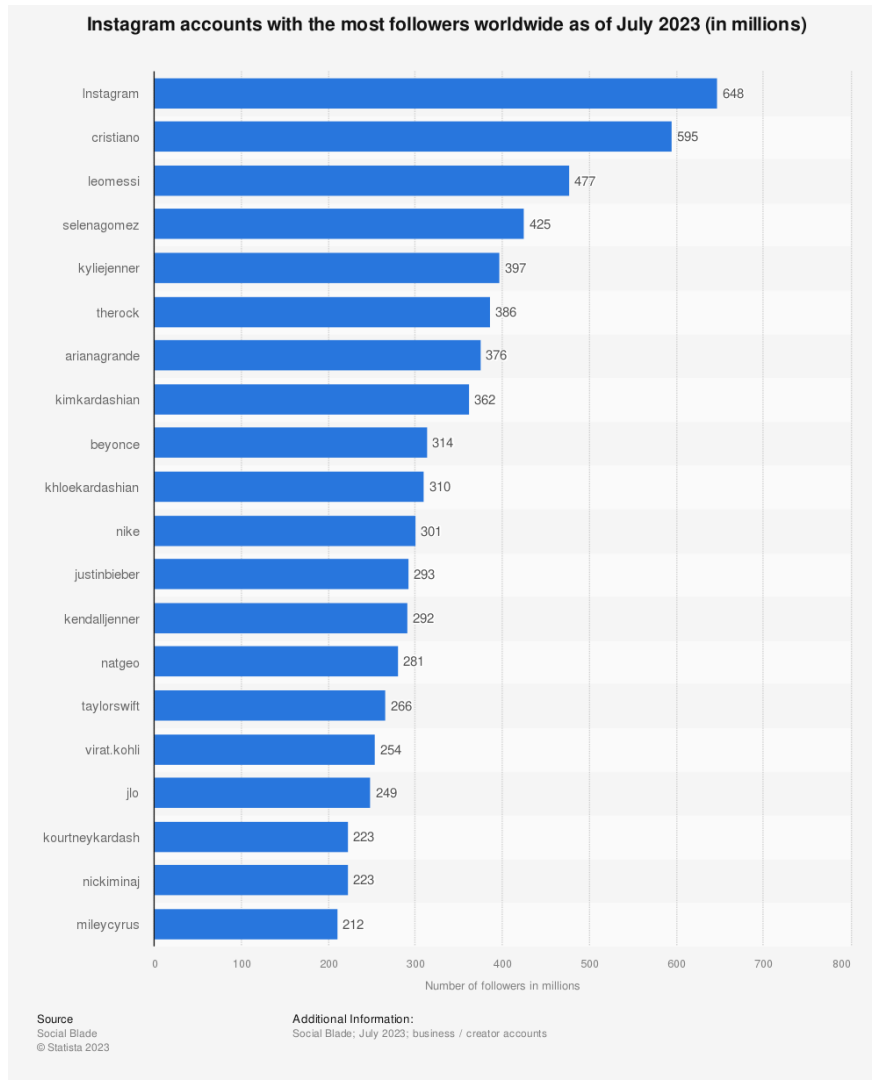
2024.



\$44B

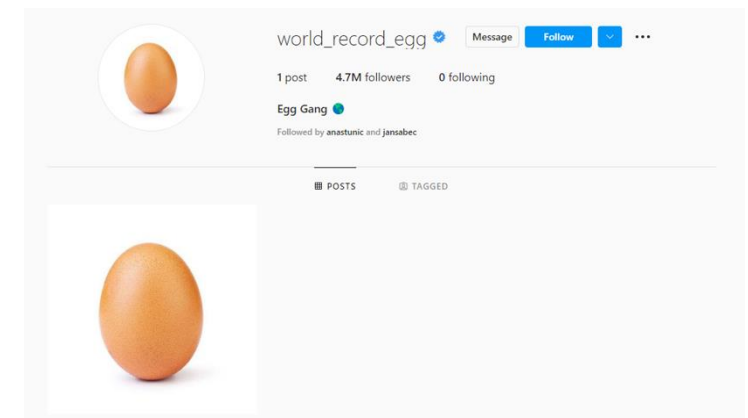
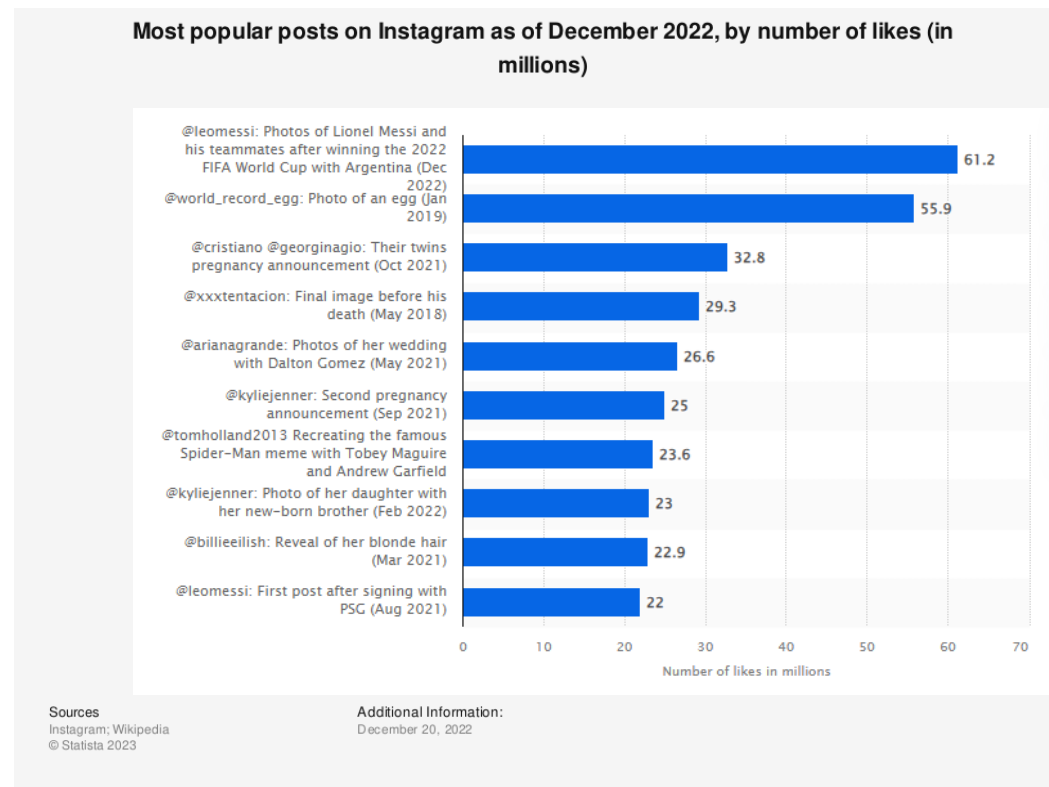
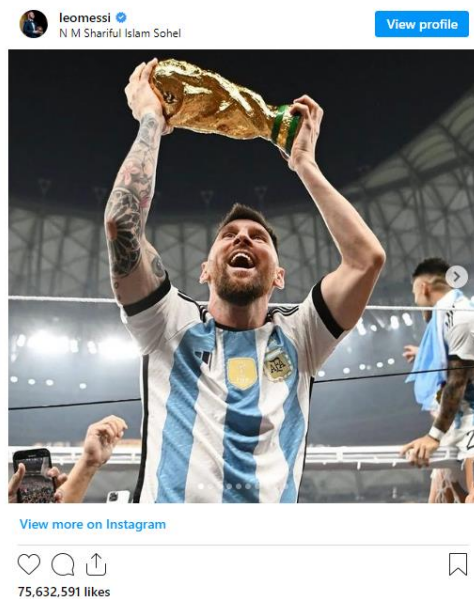


Instagram: *Najpopularniji Instagram računi*





Instagram: *Najpopularniji Instagram postovi*

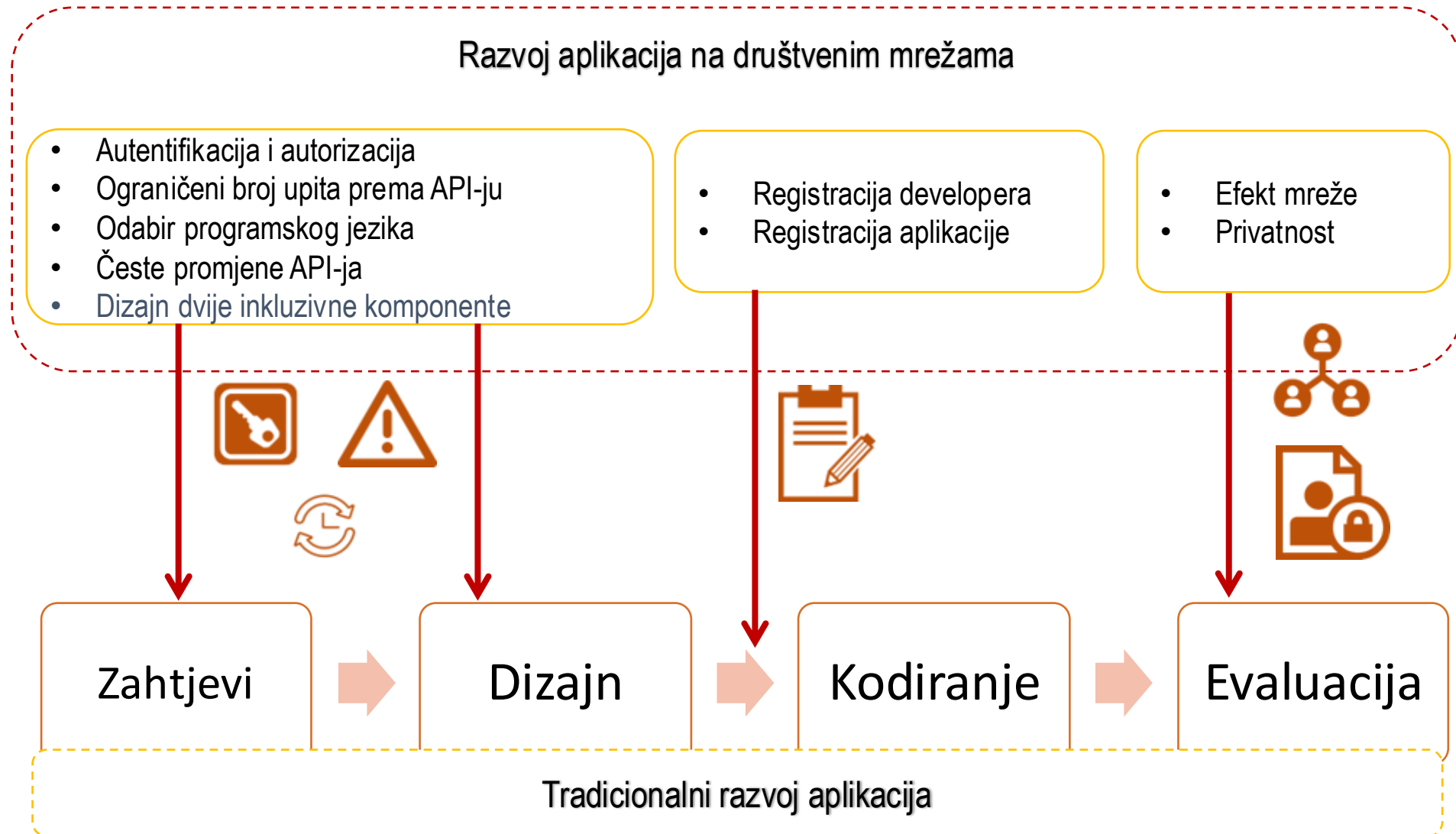


The background is a light-colored illustration of a classroom. Students are seated at desks, facing forward. On the wall, there is a clock and a network diagram consisting of several circular nodes connected by lines. Some nodes contain icons of people. The overall style is clean and modern.

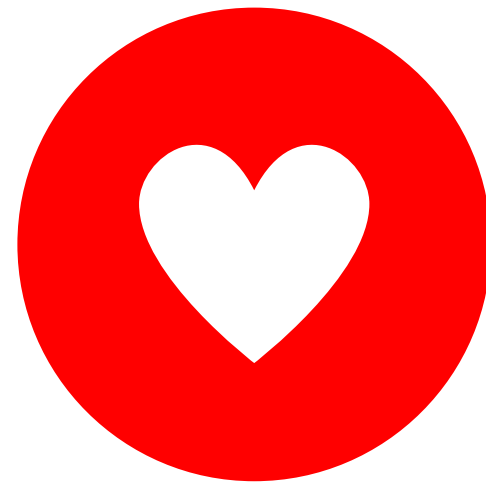
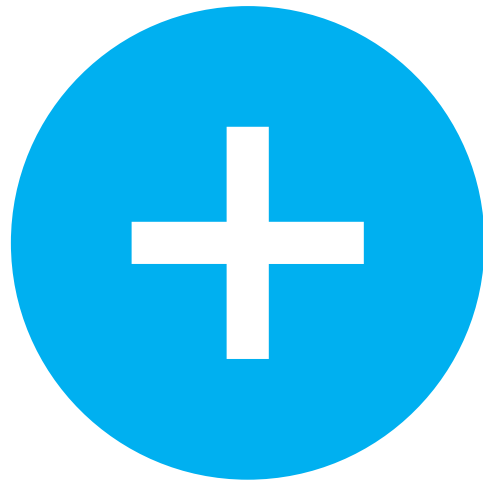
Primjeri ICT-usluga zasnovanih na društvenim mrežama



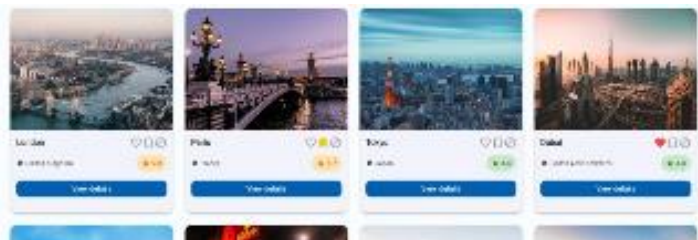
Proces razvoja aplikacija na društvenim mrežama



3 primjera studentskih rješenja



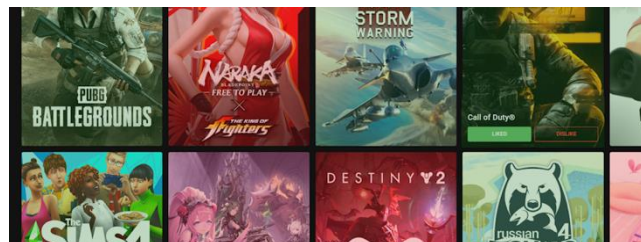
Ime projekta, autori i područje



 **Traveler**

Dona Weiner
Sara Podvorec
Filip Mohler
Dominik Papeš

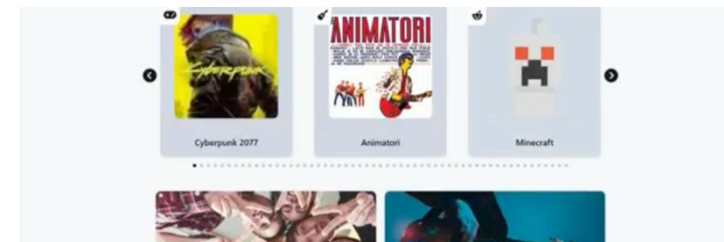
**GRADOVI
PUTOVANJA**



 **Not Just Games**

Leon Pongrac
Nikola Bukvić
Neven Marciuš

**VIDEOIGRE
FILMOVI
KNJIGE**

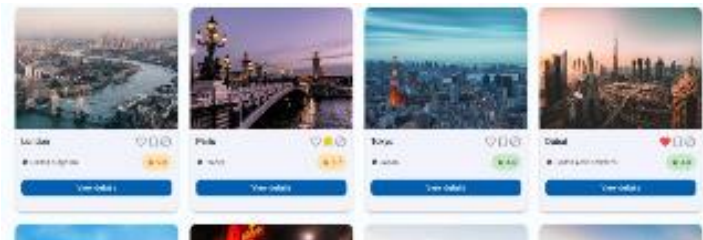


 **like.me**

Bruno Đapić
Matej Galić
Bruno Delić

**GLAZBA
VIDEOIGRE
KNJIGE**

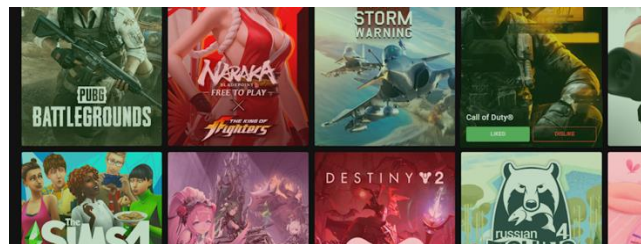
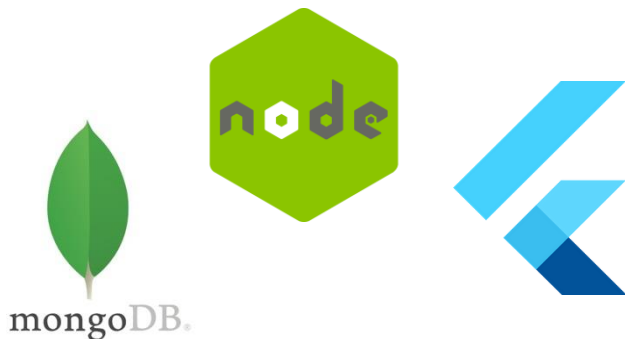
Korištene tehnologije



 **Traveler**

MongoDB

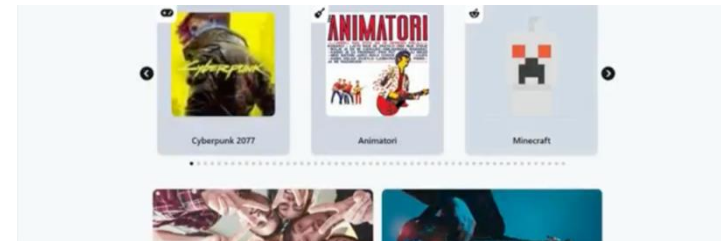
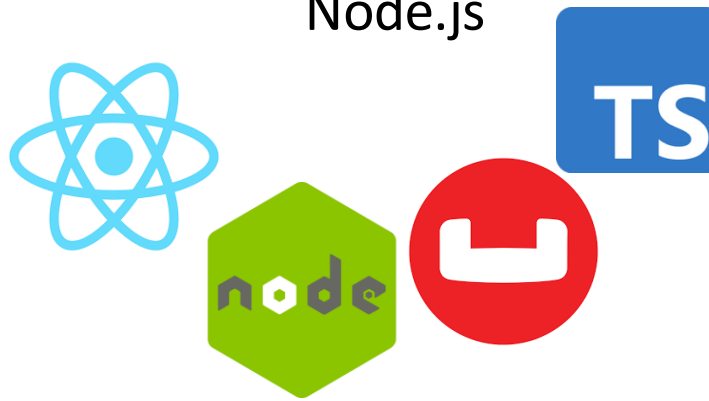
Flutter
Node.js



 **Not Just Games**

CouchBase

React
TypeScript
Node.js



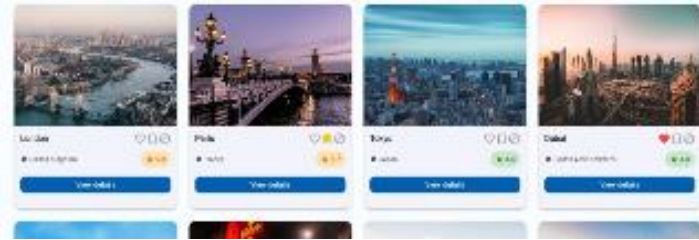
 **like.me**

MongoDB

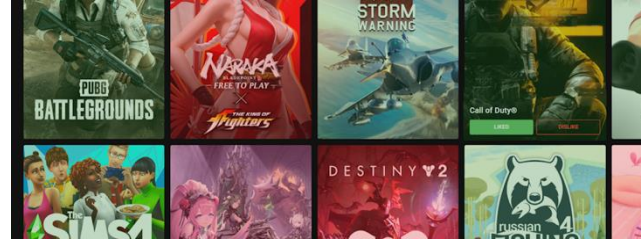
Angular2
.NET



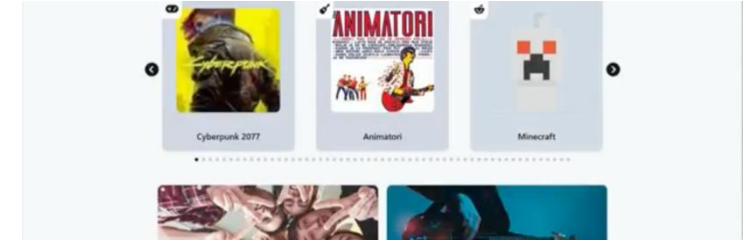
Korišteni API-ji



 **Traveler**



 **Not Just Games**



 **like.me**



Google
SerpApi
OpenWeather
Geoapify
Unsplash
Restcountries
Countries & Cities
Postman



OMDB
Steam
SteamSpy
Open Library
Google

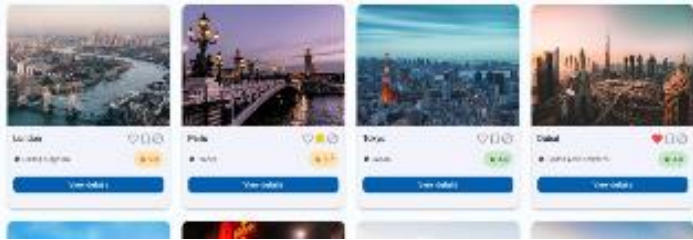


OMDb API

Spotify
Reddit
Igdb
Rainforest
GoogleAPI

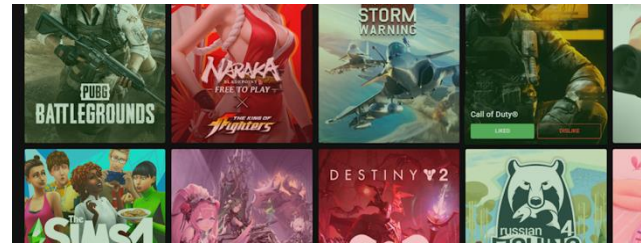


Opis projekata u brojevima



+ Traveler

- 8 korisnika
- 500+ gradova
- 50+ recenzija



★ Not Just Games

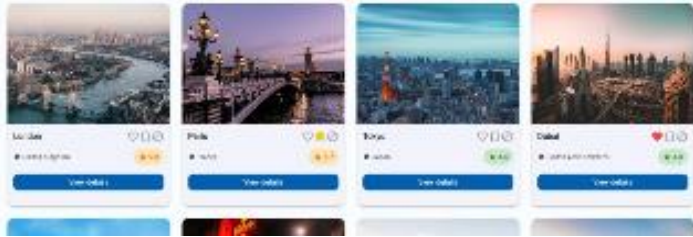
- 2000+ filmova u bazi
- 500+ knjiga
- 11000+ video igara



♥ like.me

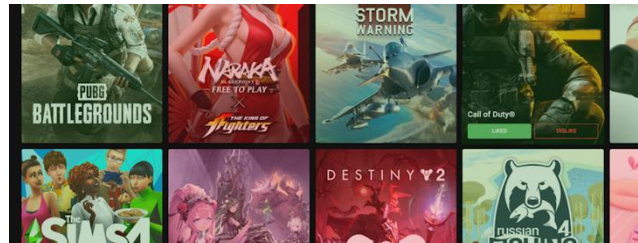
- 3 korisnika
- 15000+ knjiga
- 15000+ pjesama
- 1000+ video igara

Osnovne razvijene funkcionalnosti



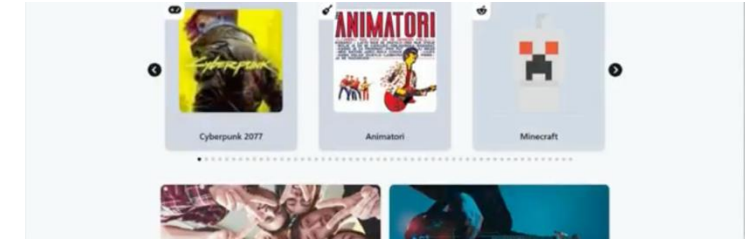
Traveler

- Prijava korisnika putem društvene mreže
- Određivanje korisnikovih preferenci na temelju onboarding upitnika
- Prikaz:
 - Informacija o gradovima
 - Popis omiljenih i željenih gradova
 - Prikaz recenzija korisnika
 - Prikaz drugih korisnika
 - Personalizirane poruke na temelju višestrukog modela
- Mogućnost komentiranja sadržaja, gradova, znamenitosti



Not Just Games

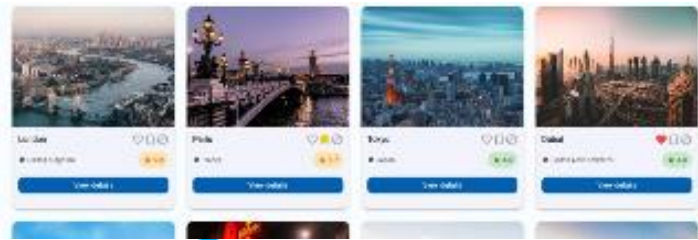
- Prijava korisnika putem društvene mreže
- Određivanje preporuka na temelju korisničkog prvotnog unosa
- Prikaz:
 - Video igara, filmova i knjiga
 - na temelju korisnikovih preferenci
 - Na temelju najigranijih igara u zadnja 2 tjedna
 - Preporuka video igara
 - Preporuka filmova i knjiga na temelju video igara



like.me

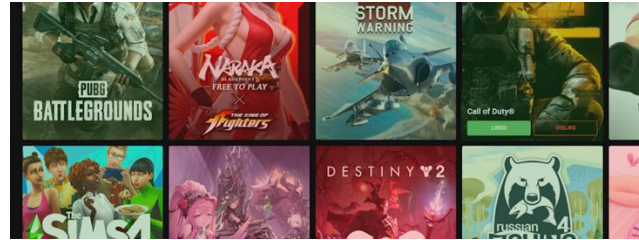
- Prijava korisnika putem više društvenih mreža
- Prikaz:
 - video igara, knjiga, subreddita, izvođača i pjesama
 - personaliziranog sadržaja obzirom na omiljene podatke i osobe koje korisnik prati
 - drugih korisnika aplikacije
- Mogućnost komentiranja sadržaja i moguće vidjeti koji korisnik sluša koju pjesmu

Preporučitelji



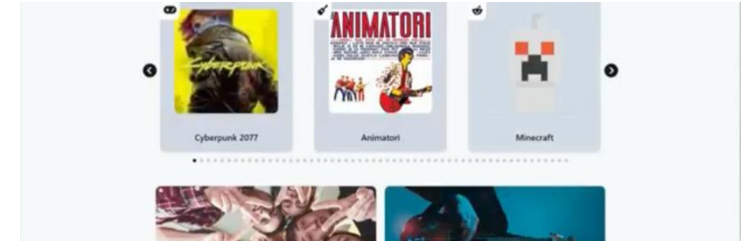
+ Traveler

- Preporuka na temelju korisnikovih preferenci iz onboarding upitnika
- Preporuka na temelju omiljenih gradova i povezanih sadržaja
- Preporuka na temelju interesa korisnikovih prijatelja



★ Not Just Games

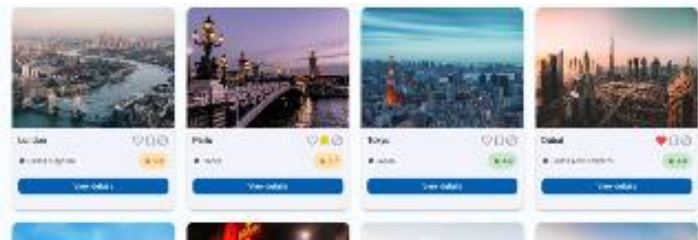
- Preporuka filmova i knjiga na temelju omiljenih video igara
- Kosinusna sličnost -> na temelju omiljenih video igara i sadržaja



♥ like.me

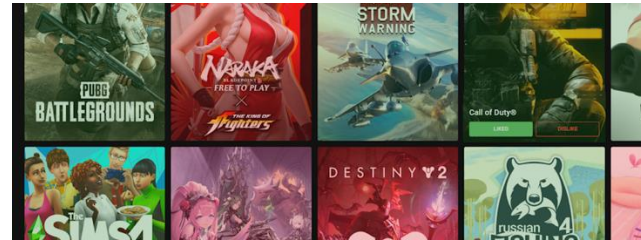
- Jaccardova sličnost
- Višekategorijska preporuka sadržaja na temelju interakcija

Međukorisničke interakcije



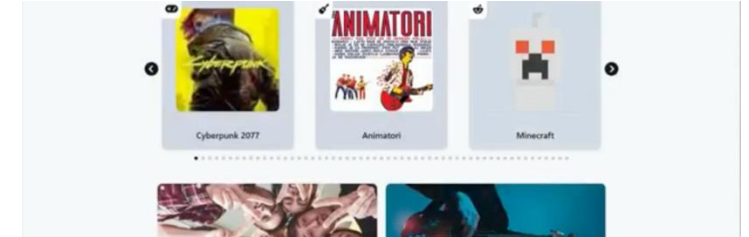
+ Traveler

- Moguće pratiti druge korisnike
- Moguće komentirati sadržaje
- Moguće međusobno komuniciranje



★ Not Just Games

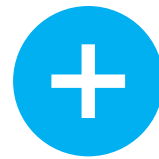
- Nema



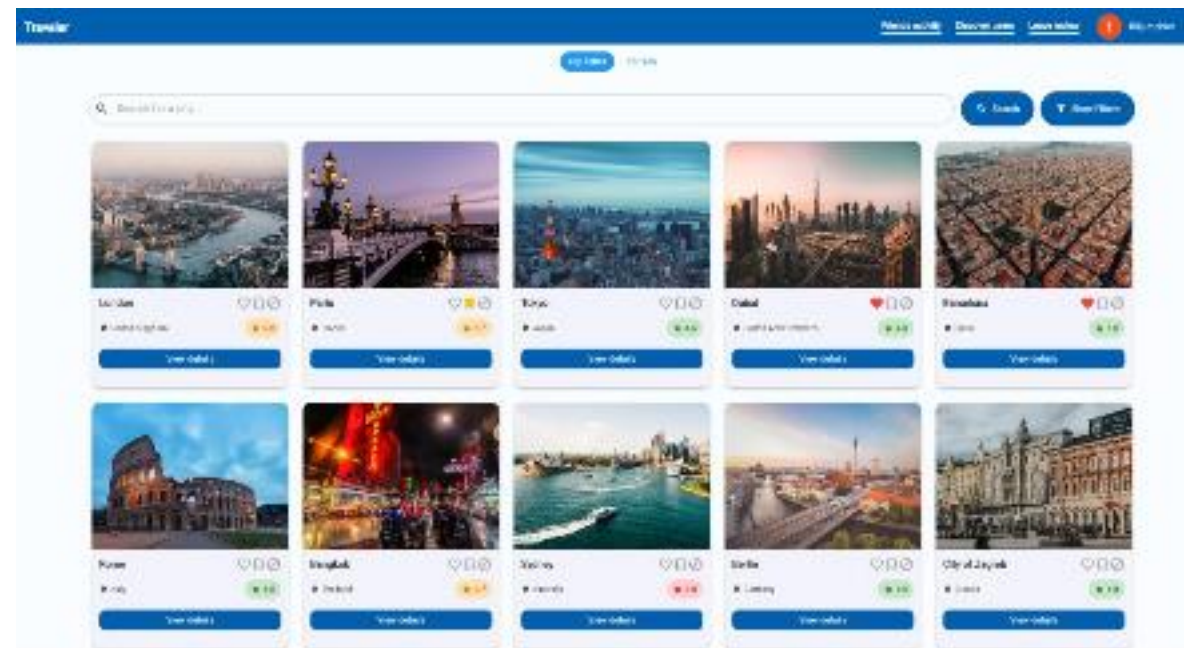
♥ like.me

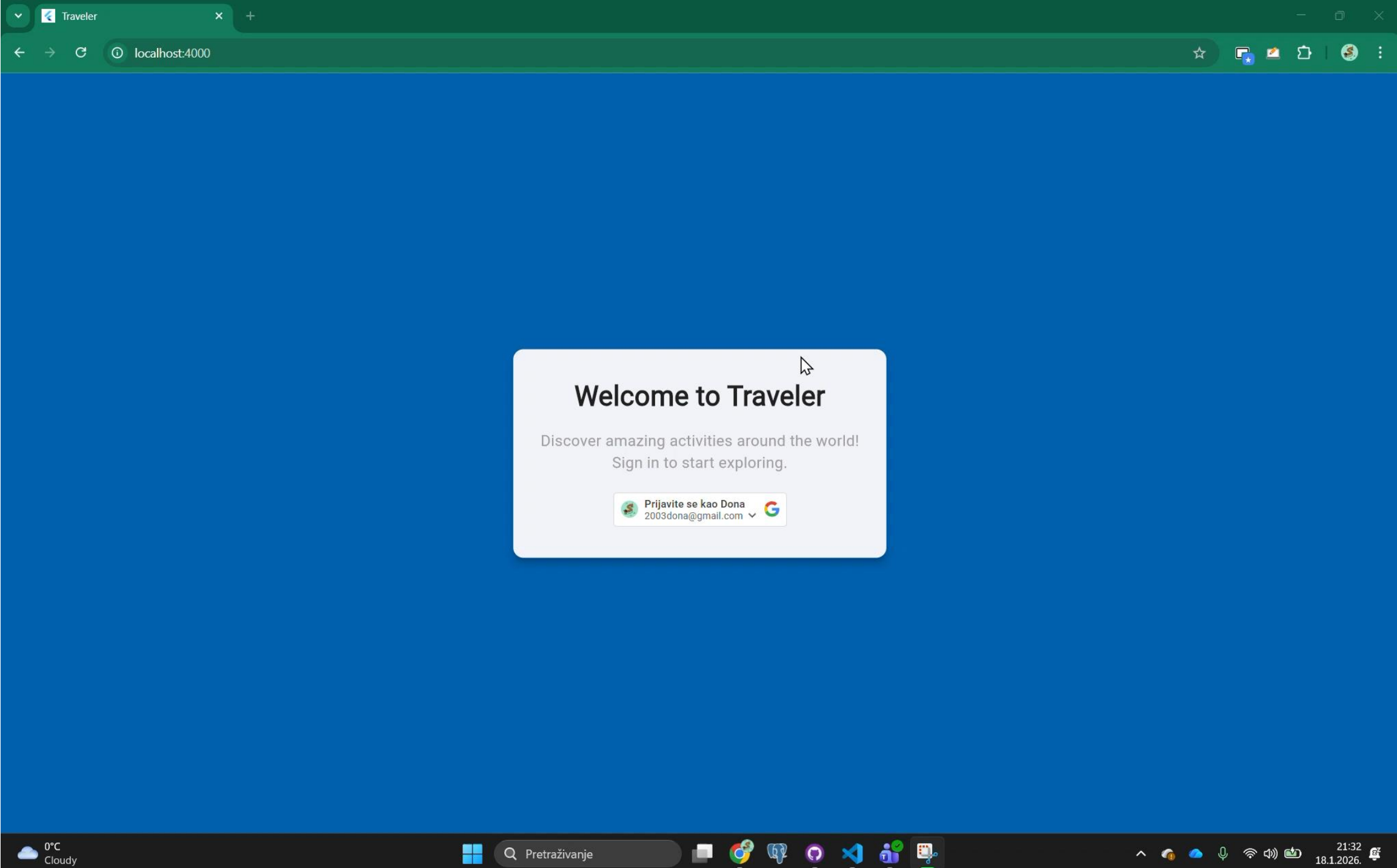
- Moguće pratiti druge korisnike
- Komentirati sadržaj
- Međusobno komunicirati

VIDEO



Traveler





Welcome to Traveler

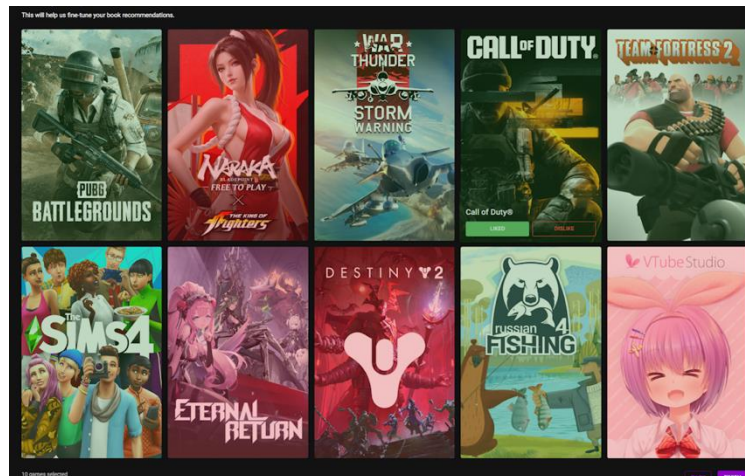
Discover amazing activities around the world!
Sign in to start exploring.

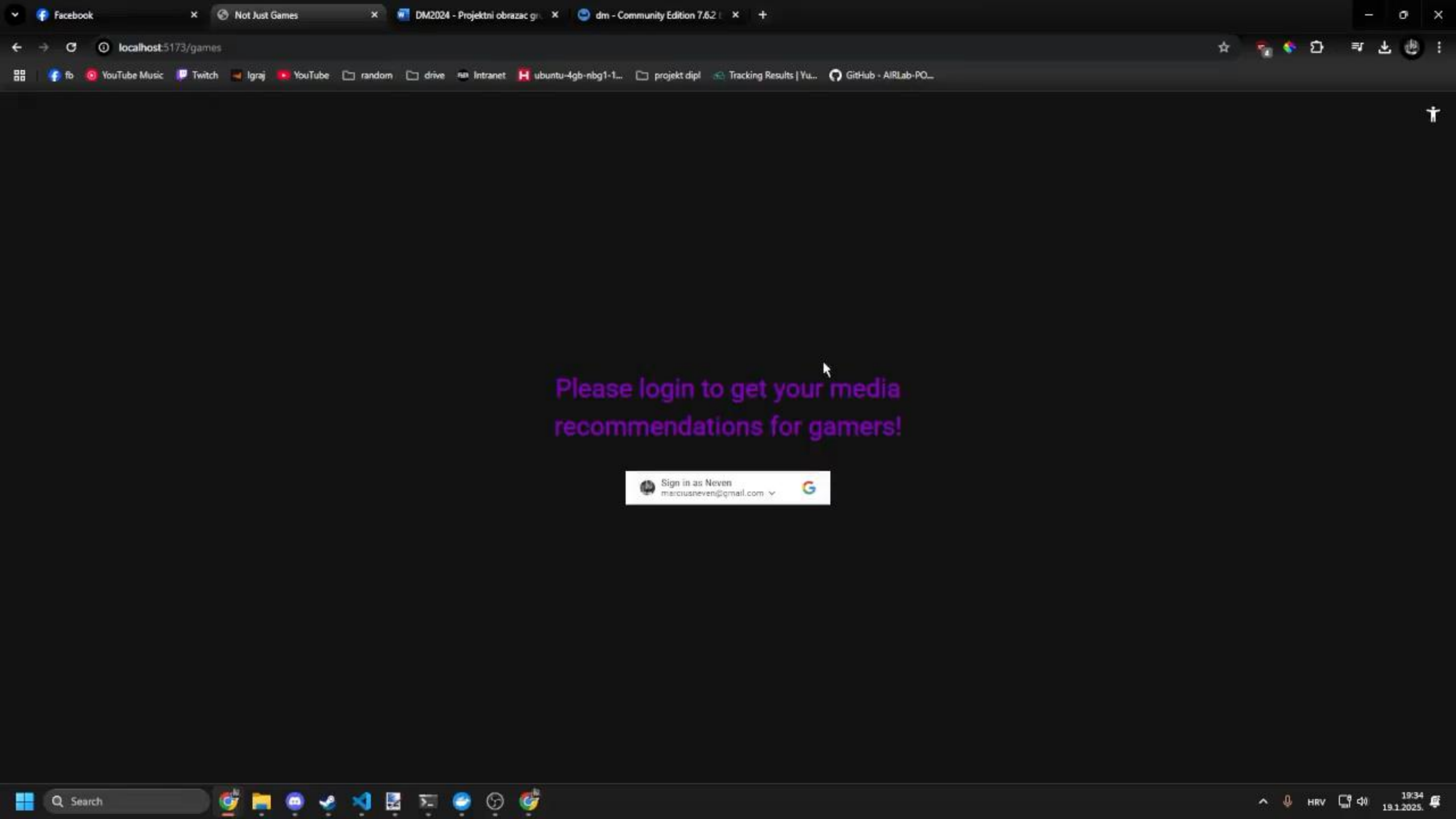
Prijavite se kao Dona
2003dona@gmail.com

VIDEO



Not Just Games





Please login to get your media
recommendations for gamers!

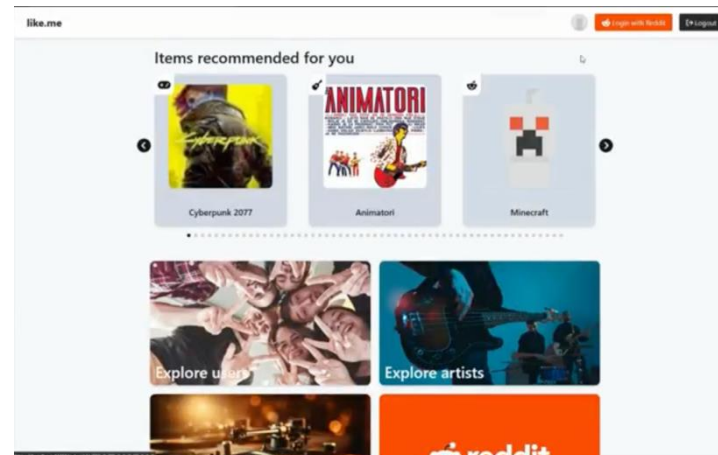
Sign in as Neven
marciusneven@gmail.com



VIDEO



like.me



Please sign in with Spotify or Reddit to find users that are like.you

Uključivi dizajn za korisnike

- UNIVERZALNI DIZAJN

- dizajn koji **odgovara svima**, može ga koristiti najviše ljudi moguće

- PRISTUPAČNOST

- osigurava da **digitalni proizvod i sadržaj** mogu koristiti ljudi s invaliditetom

- UKLJUČIVI (INKLUZIVNI) DIZAJN

- proces dizajna koji **prihvaća različitosti**, usmjeren na čovjeka

- Web Accessibility Guidelines: <https://www.w3.org/TR/WCAG21/>



Dodatno na Društvenim mrežama

- Analiza znanstvenog rada i unaprjeđenje prezentacijskih vještina
- Analiziranje metrika mreže
- Sentiment analiza sadržaja društvenih mreža

